

Data Analytics Capstone Synopsis

**Generative AI and the Pivot of Global System Integrators: Revenue Growth, AI Disclosure,
and Decline Risk in U.S.-Listed IT Services Firms, 2019–2025**

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Generative AI and the Pivot of Global System Integrators: Revenue Growth, AI Disclosure, and Decline Risk in U.S.-Listed IT Services Firms, 2019–2025

Introduction

Background and Context

Global system integrators (GSIs) such as Accenture, Cognizant, DXC Technology, Kyndryl, HCLTech, Infosys, and Wipro design, build, and run the technology estates of large enterprises. Their dominant business model is labor-based: revenue grows with the number of billable consultants, developers, and operations staff deployed on client programs. GSIs also sit at the center of the enterprise technology ecosystem. Hyperscalers, software vendors, and infrastructure manufacturers rely on GSI alliances to carry their products into large transformation and managed-services deals, so the health of a GSI directly affects the pipeline of its technology partners.

The public release of ChatGPT in November 2022 marked the start of a generative artificial intelligence (GenAI) era that directly touches the tasks GSIs sell. Eloundou et al. (2024) estimated that about 80% of U.S. workers could have at least 10% of their tasks affected by large language models (LLMs), with software and information-processing work among the most exposed. Field and experimental evidence points the same way: customer-support agents with an AI assistant resolved 15% more issues per hour (Brynjolfsson et al., 2025), professional writing tasks were completed about 40% faster with higher quality (Noy & Zhang, 2023), and developers using GitHub Copilot finished a programming task 55.8% faster (Peng et al., 2023). The McKinsey Global Institute (2023) estimated that GenAI could add US\$2.6 trillion to US\$4.4 trillion in annual value across the use cases it studied.

For GSIs this is a double-edged shift. If AI lets fewer people deliver the same scope, effort-based revenue deflates, especially in labor-intensive application maintenance, testing, and managed services. At the same time, clients need partners to design data platforms, integrate models, and govern AI in production, which creates a new services market. Firms are therefore repositioning: they announce AI practices, retrain staff, and increasingly describe AI in their

regulatory filings. Whether this repositioning is associated with better commercial outcomes is not yet clear.

The stakeholders are GSI leadership teams deciding where to invest, investors pricing the sector, enterprise buyers selecting integrators, and technology vendors' alliance teams deciding which GSI partners to prioritize for co-selling investment. The author works in this last role, managing alliances with several GSIs, which motivates the study. Only public data are used; no employer or partner-confidential information is included. Prior research links AI investment to firm growth across industries (Babina et al., 2024) but predates GenAI and does not focus on IT services. This study addresses that gap with firm-level, public, quarterly evidence from the GenAI era.

Problem Statement

IT services firms and GSIs are repositioning around generative AI, but it is not known whether the GenAI era has changed their revenue trajectories, whether firms that emphasize AI in their disclosures grow faster, which firms are at risk of revenue decline, or which service business models are most exposed. Decisions by GSI leaders, investors, and technology-partner teams are therefore being made on narrative rather than evidence. This study measures these relationships on a panel of U.S.-listed IT services firms (SEC Standard Industrial Classification [SIC] codes 7370, 7371, 7373, and 7374) from 2019 to 2025. The **dependent variables** are year-over-year revenue growth (REV_GROWTH_YOY, percent) and revenue-decline status (DECLINE_Q and next-quarter DECLINE_NEXT_Q, binary). The **independent variables** are the GenAI period (POST_GENAI), AI-disclosure intensity (AI_INTENSITY, AI mentions per 10,000 words of 10-Q/10-K text), and service segment (SEGMENT). Firm size, SG&A intensity, operating margin, and prior growth act as controls. To test whether any shift is specific to U.S.-listed firms, 15 IT services firms listed on the National Stock Exchange of India (NSE) form a comparison group (MARKET).

Purpose of the Study

The study has two aims. First, it *explains drivers*: it tests whether growth differs between the pre-GenAI and post-GenAI periods, whether AI-disclosure intensity is associated with growth after controls, and whether decline risk differs by service segment. Second, it *predicts*: it builds and evaluates a classification model that flags firms likely to report a revenue decline in the next quarter. Together these produce an evidence-based early-warning view of GSI health that practitioners can refresh every quarter from public filings.

Scope and Objectives

Project Scope and Boundaries

The unit of analysis is the **firm-quarter**. The population is U.S. Securities and Exchange Commission (SEC) registrants that file 10-Q and 10-K reports with XBRL financial data and are classified under SIC 7370, 7371 (computer programming services), 7373 (computer integrated systems design), or 7374 (computer processing and data preparation) (SEC, n.d.-c). GSIs whose SIC code lies outside this range, such as Accenture, are added by name (Appendix A). Code 7372 (prepackaged software) is excluded because those firms sell products rather than services. The study window covers calendar quarters 2019Q1 to 2025Q4, with 2018 used only to compute year-over-year growth for 2019. Registrants with trailing-twelve-month revenue below US\$10 million are removed to exclude shells.

Three segments are formed from SIC codes: *IT consulting and application services* (7370/7371), *systems integration* (7373), and *data processing and managed services* (7374). The pre-GenAI period is 2019Q1–2022Q3, the post-GenAI period is 2023Q1–2025Q4, and 2022Q4 is treated as a transition quarter and excluded from period comparisons.

The scope deliberately excludes: (a) causal claims, since the design is observational; (b) share prices and valuation; (c) headcount, which is not consistently tagged in XBRL; (d) foreign private issuers that file only annual 20-F reports with the SEC (India-listed firms are instead covered through the NSE comparison group described below); and (e) any confidential deal,

pipeline, or partner data. These boundaries keep the project feasible within 10 weeks while still covering a sector-wide panel expected to contain several thousand firm-quarters.

Indian comparison group. Several of the largest GSIs (for example, TCS, Infosys, HCLTech, and Wipro) report quarterly results to the NSE rather than filing 10-Q reports with the SEC. To include them, the study adds a comparison group of 15 NSE-listed IT services firms: TCS, Infosys, HCLTech, Wipro, Tech Mahindra, LTIMindtree, Mphasis, Coforge, Persistent Systems, Birlasoft, Zensar Technologies, Mastek, Sonata Software, Cyient, and L&T Technology Services. Their consolidated quarterly results are read from the XBRL files each company files with the NSE under Indian Accounting Standards (Ind AS). These firms do not file 10-Q or 10-K text, so they enter RQ1 as a second market and a robustness check but not the AI-disclosure analyses (RQ2 and RQ3) or RQ4. They add about 340 firm-quarters to the 2019–2025 window and are not counted toward the U.S. sample-size requirement.

Research Questions and Hypotheses

The four research questions are facets of one topic, and each is paired with a null (H_0) and alternative (H_1) hypothesis tested at $\alpha = .05$.

RQ1. Is the mean year-over-year revenue growth of U.S.-listed IT services firms in the post-GenAI period (2023Q1–2025Q4) different from that in the pre-GenAI period (2019Q1–2022Q3)?

$H_{01}: \mu_{\text{post}} = \mu_{\text{pre}}$, $H_{11}: \mu_{\text{post}} \neq \mu_{\text{pre}}$. The same test is repeated for the Indian comparison group.

RQ2. In the post-GenAI period, is AI-disclosure intensity associated with year-over-year revenue growth after controlling for firm size, SG&A ratio, operating margin, prior growth, and segment?

$H_{02}: \beta_{\text{AI}} = 0$, $H_{12}: \beta_{\text{AI}} \neq 0$.

RQ3. Can financial indicators combined with AI-disclosure features predict which firms will report a revenue decline in the next quarter better than chance?

H_{03} : AUC = 0.50 (no discrimination). H_{13} : AUC > 0.50. The practical target is AUC $\geq$ 0.70 on a held-out 2025 test set.

RQ4. In the post-GenAI period, is revenue-decline status associated with service segment?

H_{04} : DECLINE_Q and SEGMENT are independent. H_{14} : DECLINE_Q and SEGMENT are associated.

Variables

Table 1 summarizes the analytical variables, their measurement, and their role in each research question. Full definitions, including identifiers, appear in the data dictionary (Table 4). All variables are measurable from public SEC and NSE data.

Table 1

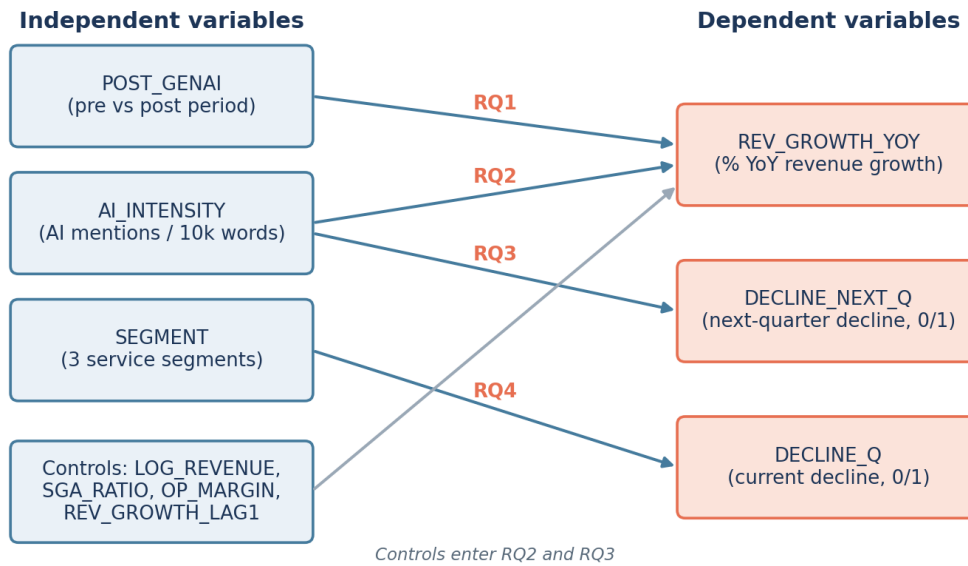
Analytical Variables, Types, Units, and Roles

Variable	Type	Unit	Role
REV_GROWTH_YOY	Continuous	Percent	DV (RQ1, RQ2); feature (RQ3)
DECLINE_NEXT_Q	Binary (0/1)	None	DV (RQ3)
DECLINE_Q	Binary (0/1)	None	DV (RQ4)
POST_GENAI	Binary (0/1)	None	IV (RQ1); feature (RQ3)
AI_INTENSITY	Continuous	Mentions per 10,000 words	IV (RQ2); feature (RQ3)
GENAI_MENTION	Binary (0/1)	None	Feature (RQ3)
SEGMENT	Nominal, 3 levels	None	IV (RQ4); control (RQ2, RQ3)
MARKET	Nominal, 2 levels	None	Moderator (RQ1 robustness)
LOG_REVENUE	Continuous	ln(US\$)	Control (RQ2); feature (RQ3)
SGA_RATIO	Continuous	Percent of revenue	Control (RQ2); feature (RQ3)
OP_MARGIN	Continuous	Percent of revenue	Control (RQ2); feature (RQ3)
REV_GROWTH_LAG1	Continuous	Percent	Control (RQ2); feature (RQ3)

Note. DV = dependent variable; IV = independent variable. Ratio variables are winsorized at the 1st and 99th percentiles.

Figure 1

Conceptual Framework Linking Independent and Dependent Variables to RQ1–RQ4



Note. Arrows show the hypothesized relationships tested in each research question. Controls enter the RQ2 regression and the RQ3 classifiers.

Sample Size Calculation

Each research question uses the sample-size method that matches its analysis.

Assumptions are held constant across questions: $\alpha = .05$ (two-tailed where applicable), power ($1 - \beta$) = .80, and Cohen's (1988) conventions for a *small* effect, chosen because GenAI effects on sector revenue are likely to be modest. Calculations follow Cohen (1988) and Faul et al. (2007) and are reproduced in `src/00_sample_size.py` in the project repository.

RQ1: Power Analysis for Two Independent Means

RQ1 compares mean growth between two periods with a Welch two-sample t-test, so an a priori power analysis for a difference in means is appropriate. With effect size $d = 0.20$, $z_{1-\alpha/2} = 1.960$, and $z_{1-\beta} = 0.842$:

$$n \text{ per group} = 2(z_{1-\alpha/2} + z_{1-\beta})^2 / d^2 = 2(1.960 + 0.842)^2 / 0.20^2 = 2(7.849) / 0.04 = 392.5$$

Correcting for the t distribution gives 393.4, which rounds up to **394 firm-quarters per period**, so RQ1 needs **N = 788**.

RQ2: Power Analysis for a Regression Coefficient

RQ2 tests one coefficient (β_{AI}) in a multiple regression with $p = 7$ predictors (AI_INTENSITY, LOG_REVENUE, SGA_RATIO, OP_MARGIN, REV_GROWTH_LAG1, and two SEGMENT dummies), so the F-test power method applies. With Cohen's $f^2 = 0.02$ (small), numerator $df = 1$, and denominator $df = N - p - 1$, the noncentrality parameter is $\lambda = f^2 \times N$. Power reaches .80 when $\lambda \approx 7.89$, so $N \approx 7.89 / 0.02 = 394.4$. Solving iteratively with the noncentral F distribution gives **N = 395** (achieved power = .801).

RQ3: Confidence-Interval Method for a Proportion

RQ3 is a classification task. Closed-form power formulas for machine-learning classifiers are not well defined, so the confidence-interval method is used. The estimate that drives the design is the prevalence of the positive class (the proportion of firm-quarters followed by a revenue decline), because it sets class balance, threshold choice, and the baseline for precision and recall. Using the most conservative prevalence $p = 0.50$, a margin of error $e = \pm 0.03$, and $Z = 1.96$:

$$n = Z^2 \times p(1 - p) / e^2 = 1.96^2 \times 0.50 \times 0.50 / 0.03^2 = 3.8416 \times 0.25 / 0.0009 = 1,067.1$$

This rounds up to **N = 1,068**. As a cross-check, the events-per-variable rule of at least 10 events per predictor (Peduzzi et al., 1996), with 10 predictors and an expected decline prevalence of 30%, requires $10 \times 10 / 0.30 = 334$ firm-quarters. The CI requirement is stricter and is adopted.

RQ4: Power Analysis for a Chi-Square Test of Independence

RQ4 uses a 3 (segment) $\times$ 2 (decline status) chi-square test, so $df = (3 - 1)(2 - 1) = 2$. With Cohen's $w = 0.10$ (small), power .80 at $\alpha = .05$ requires a noncentrality parameter $\lambda = 9.635$:

$$N = \lambda / w^2 = 9.635 / 0.10^2 = 963.5$$

This rounds up to **N = 964**.

Table 2*Sample Size Calculation for RQ1–RQ4*

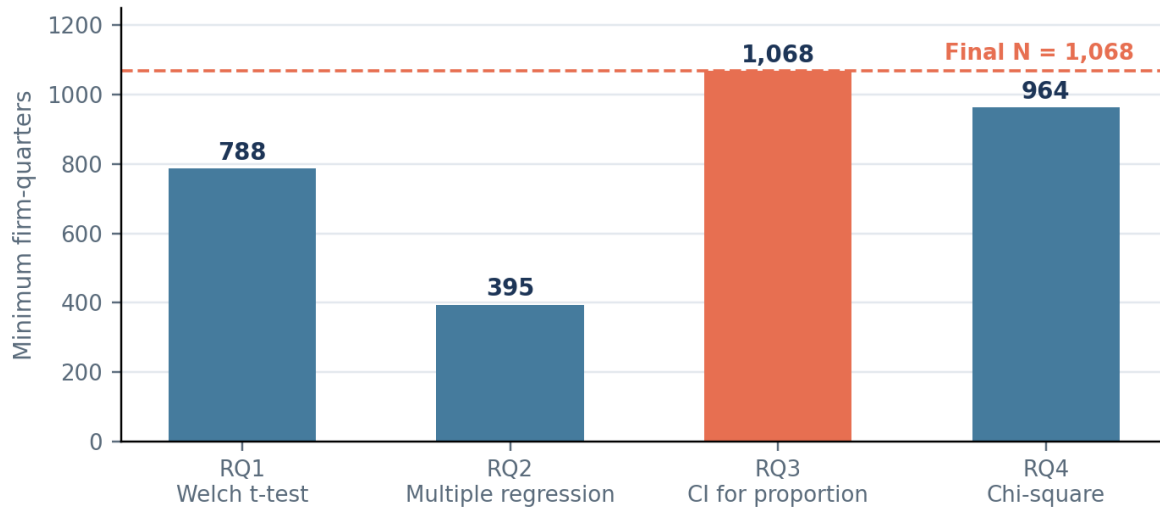
Research question	Method used	Key parameters	Minimum sample size (N)
RQ1: Growth pre vs post GenAI	Power (two-sample t-test)	$\alpha = .05$, power = .80, $d = 0.20$ (394 per group)	788
RQ2: AI intensity and growth	Power (multiple regression F-test)	$\alpha = .05$, power = .80, $f^2 = 0.02$, 1 tested of 7 predictors	395
RQ3: Predict next-quarter decline	Confidence interval (proportion)	$Z = 1.96$ (95%), $e = 0.03$, $p = 0.50$	1,068
RQ4: Segment and decline	Power (chi-square, $df = 2$)	$\alpha = .05$, power = .80, $w = 0.10$	964
Final sample size	Maximum across RQ1–RQ4	$\max(788, 395, 1,068, 964)$	1,068

Note. Units are firm-quarters. Values were computed with SciPy and statsmodels and match G*Power (Faul et al., 2007).

Final sample size decision. To keep adequate power across all analyses, the study adopts the maximum of the four minimum sample sizes: **N = 1,068 firm-quarters**. This minimum applies to every analysis sample: the full 2019–2025 panel for RQ1 and RQ3, and the post-GenAI subsample for RQ2 and RQ4. Because repeated quarters from the same firm are correlated, all inferential tests will use firm-clustered standard errors (Petersen, 2009). The design effect $DEFF = 1 + (m - 1) \times ICC$ (Gelman & Hill, 2007), where m is quarters per firm and ICC is the intraclass correlation, will be reported. If the available panel is smaller than $1,068 \times DEFF$, the SIC range or time window will be widened before analysis begins. The full panel will be used, since a larger sample only increases precision.

Figure 2

Minimum Sample Size Required by Each Research Question



Note. The highest requirement (RQ3) sets the final sample size of 1,068 firm-quarters.

Data Description

Data Sources and Access

All data are free, public, and official. U.S. data come from the SEC's EDGAR system, Indian data from the NSE's corporate filings, and exchange rates from the Federal Reserve Bank of St. Louis. No Kaggle or synthetic data are used. The sources are listed in Table 3. EDGAR's application programming interfaces (APIs) need no key, but automated users must send a descriptive User-Agent header with contact details and stay under 10 requests per second (SEC, n.d.-a). The project scripts enforce both rules and cache every download so that results can be reproduced. The NSE site often blocks automated requests from cloud servers, so a dated snapshot of the 488 parsed NSE filings and the exchange-rate series is committed to the repository; the pipeline tries the live NSE feeds first and falls back to the snapshot.

Table 3*Data Sources*

Source	Content used	URL
EDGAR submissions API	Firm SIC code, filing index (form, report date, accession number)	https://data.sec.gov/submissions/CIK#####.json
EDGAR XBRL company facts API	Quarterly and annual revenue, operating income, SG&A (US-GAAP tags)	https://data.sec.gov/api/xbrl/companyfacts/CIK#####.json
EDGAR filing archives	Full text of 10-Q and 10-K primary documents for AI keyword scoring	https://www.sec.gov/Archives/edgar/data/
SEC SIC code list	Industry definitions for SIC 7370–7374	https://www.sec.gov/search-filings/standard-industrial-classification-sic-code-list
NSE corporate filings (financial results and integrated filings)	Consolidated quarterly Ind AS results in XBRL for the 15 Indian comparison firms	https://www.nseindia.com/companies-listing/corporate-filings-financial-results
FRED series DEXINUS	Quarterly average Indian rupees per U.S. dollar	https://fred.stlouisfed.org/series/DEXINUS

Note. API documentation: SEC (n.d.-b). ##### denotes the firm's 10-digit Central Index Key.

Quarterly financial values are taken from XBRL facts covering about 90 days. Where a firm reports only a fiscal-year total for its fourth quarter, Q4 is derived as the annual total minus the three reported quarters. Fiscal quarters are mapped to calendar quarters so that firms with different fiscal years can be compared. For the Indian firms, revenue is Ind AS *RevenueFromOperations*, and operating income is profit before tax plus finance costs minus other income. Their growth is computed in rupees, so that exchange-rate movements do not distort it, while revenue levels are converted to U.S. dollars at the quarterly average DEXINUS rate (Board of Governors of the Federal Reserve System, n.d.). AI-disclosure intensity is computed from filing text with a transparent keyword dictionary (Appendix A). Following Loughran and McDonald (2011), the dictionary is built for financial text rather than taken from a general-purpose word list, and its precision is checked by manually reviewing a random sample of 50 matched passages.

Ethical Considerations

The data describe companies, not individuals, and contain no personal information. SEC materials are public-domain government records, and the SEC's fair-access policy is respected. The author has a professional relationship with several GSIs. To manage this conflict of interest, no proprietary, partner-confidential, or employer data are used, and every firm is processed by the same code and rules. Firm-level results will be discussed at sector and segment level rather than singling out individual partners. The study also guards against several forms of bias. Firms that deregistered during the window are retained to avoid survivorship bias. Model features use only information available at the end of each quarter, and the train, validation, and test splits are time-ordered to prevent look-ahead bias. Keyword counts may reward promotional "AI washing" rather than real capability, so AI_INTENSITY is interpreted as *disclosed emphasis*, not verified AI revenue.

Limitations

First, the design is observational, so associations cannot be read as causal effects. The post-GenAI period also coincides with interest-rate rises and a post-pandemic normalization of IT spending. Second, the Indian comparison group is small (15 firms), reports under Ind AS rather than US GAAP, has no separate SG&A line, and has no 10-Q/10-K text, so it supports the RQ1 comparison only and cannot test the AI-disclosure questions. Third, XBRL tagging varies across firms (for example, *Revenues* versus *RevenueFromContractWithCustomerExcludingAssessedTax*), which the extraction code reconciles but cannot perfectly standardize. Fourth, keyword-based text measures carry measurement error, and SIC codes are coarse, self-reported proxies for business model. Finally, the 2025 test quarters may reflect conditions that do not persist.

Data Dictionary

Table 4 defines every variable in the final analysis file, *data/processed/panel_firm_quarter.csv*. The same dictionary is stored as *docs/data_dictionary.csv* in the repository.

Table 4*Data Dictionary for the Firm-Quarter Panel*

Variable	Type	Unit	Definition and source	Relevance
cik	Nominal	None	SEC Central Index Key; NSE:symbol for Indian firms	Firm ID; cluster variable
ticker	Nominal	None	Exchange ticker (SEC company tickers file)	Readability; named GSIs
name	Nominal	None	Registrant name	Readability
MARKET	Nominal (2 levels)	None	US = SEC registrant; India = NSE comparison firm	Market comparison (RQ1)
sic	Nominal	4-digit code	SEC SIC code (blank for Indian firms)	Defines universe and segment
SEGMENT	Nominal (3 levels)	None	IT consulting & application services (7370/7371); systems integration (7373); data processing & managed services (7374)	Business-model exposure (RQ4)
FISCAL_QTR	Ordinal	Calendar quarter	Fiscal quarter-end mapped to calendar quarter (e.g., 2024Q2)	Time index
PERIOD	Nominal (3 levels)	None	pre = 2019Q1–2022Q3; transition = 2022Q4; post = 2023Q1–2025Q4	RQ1 groups
POST_GENAI	Binary	None	1 if PERIOD = post	GenAI era (RQ1, RQ3)
REVENUE_USD	Continuous	US\$	Quarterly revenue from XBRL; Indian revenue converted at DEXINUS	Basis for growth and size
LOG_REVENUE	Continuous	ln(US\$)	Natural log of REVENUE_USD	Size control
REV_GROWTH_YOY	Continuous	Percent	$(\text{Rev}_t - \text{Rev}_{t-4}) / \text{Rev}_{t-4} \times 100$, in reporting currency	Primary outcome (RQ1, RQ2)
REV_GROWTH_LAG1	Continuous	Percent	REV_GROWTH_YOY in quarter $t - 1$	Momentum control
DECLINE_Q	Binary	None	1 if REV_GROWTH_YOY < 0	Outcome (RQ4)
DECLINE_NEXT_Q	Binary	None	DECLINE_Q in quarter $t + 1$	Outcome (RQ3)
OP_MARGIN	Continuous	Percent	Operating income / revenue $\times 100$	Profitability control
SGA_RATIO	Continuous	Percent	SG&A expense / revenue $\times 100$ (U.S. only)	Cost-structure control
AI_INTENSITY	Continuous	Per 10,000 words	AI keyword matches in 10-Q/10-K text / words $\times 10,000$	AI positioning (RQ2, RQ3)
GENAI_MENTION	Binary	None	1 if any GenAI-specific term appears	GenAI positioning (RQ3)

Note. SG&A = selling, general, and administrative expense. Ratios are winsorized at the 1st and 99th percentiles.

Data Repository

All code, configuration, the data dictionary, the NSE and exchange-rate snapshots, and instructions are kept in the project GitHub repository: <https://github.com/ckt-github/gsi-genai-capstone>. The README explains how to install the requirements, set the SEC User-Agent, and run the numbered scripts in order to rebuild the panel from scratch. Two GitHub Actions workflows automate this. The first checks every change: the scripts compile, unit tests pass, the sample-size calculation reproduces $N = 1,068$, and the Indian data build correctly. The second runs the full pipeline in the cloud on demand and on the 2nd of each month, and commits the resulting interim files, analysis panel, and results back to the repository. The text-scoring step saves progress after each firm, so an interrupted run resumes where it stopped. Figure 3 shows the repository structure.

Figure 3*Structure of the Project GitHub Repository*

gsi-genai-capstone/	
├ README.md	setup and run instructions
├ requirements.txt	Python packages
├ .github/workflows/	
│ └ ci.yml	checks on every change
└ pipeline.yml	cloud run; commits outputs
├ config/	
│ └ settings.yaml	period, SIC codes, filters
│ └ ai_keywords.txt	AI / GenAI term dictionary
│ └ named_gsis.csv	U.S.-listed GSIs added by name
└ india_firms.csv	15 NSE comparison firms
├ data/	
│ └ raw/india/	NSE XBRL and FX snapshots
│ └ interim/	parsed financials, text scores
│ └ processed/	
│ └ └ panel_firm_quarter.csv	
├ docs/	data_dictionary.csv, synopsis.pdf
├ results/	analysis output
├ tests/	unit tests
└ src/	
│ └ 00_sample_size.py	RQ1-RQ4 sample sizes
│ └ 01_build_universe.py	SEC firms by SIC code
│ └ 02_get_financials.py	U.S. XBRL financials
│ └ 02b_india_financials.py	NSE financials, INR-USD
│ └ 03_ai_intensity.py	10-Q/10-K AI scores
│ └ 04_build_panel.py	merge and derive variables
└ 05_analysis.py	RQ1-RQ4 tests and models

Note. Scripts are numbered in execution order. Raw SEC downloads in *data/raw* are rebuilt by the scripts and are not committed; *data/raw/india* holds the committed NSE and exchange-rate snapshots.

Analytic Approach

Each research question is mapped to one primary method, with robustness checks. Table 5 summarizes the mapping, assumptions, and outputs; the subsections give the details and evaluation formulas.

Table 5*Mapping of Research Questions to Analytic Methods*

RQ	Primary method	Key assumptions and checks	Outputs
RQ1	Welch two-sample t-test; robustness: Mann–Whitney U, firm-level paired t-test, excluding 2020, POST × MARKET regression with Indian firms	Independence handled by firm clustering; unequal variances allowed; large-n normality of means	Mean difference with 95% CI, t, p, Cohen's d
RQ2	OLS multiple regression with firm-clustered standard errors; robustness: quarter fixed effects	Linearity, homoscedasticity (Breusch–Pagan), VIF < 5, residual normality	β_{AI} with 95% CI and p; R ² , adjusted R ² , RMSE, MAE
RQ3	Logistic regression, random forest, XGBoost; time-ordered split	No look-ahead; class weights for imbalance; threshold tuned on validation	ROC-AUC with DeLong CI, accuracy, precision, recall, F1, Brier score, SHAP importance
RQ4	Chi-square test of independence; follow-up logistic regression	Expected counts ≥ 5; independent observations (checked by firm-level sensitivity)	χ^2 , p, Cramér's V, standardized residuals

Solution to RQ1

RQ1 is answered with statistical inference. The Welch t-test compares mean REV_GROWTH_YOY between the pre (2019Q1–2022Q3) and post (2023Q1–2025Q4) periods without assuming equal variances:

$$t = (\bar{x}_{\text{post}} - \bar{x}_{\text{pre}}) / \sqrt{(s^2_{\text{post}} / n_{\text{post}} + s^2_{\text{pre}} / n_{\text{pre}})}$$

Effect size is reported as Cohen's d = $(\bar{x}_{\text{post}} - \bar{x}_{\text{pre}}) / s_{\text{pooled}}$. Because 2020 contains pandemic shocks, the test is repeated excluding 2020. Because firm-quarters are clustered, it is also repeated on firm-level period means with a paired t-test, and with a Mann–Whitney U test in case growth is heavily skewed. H_{01} is rejected if $p < .05$. The output is the change in average sector growth with its 95% confidence interval and effect size. The test is then repeated for the Indian comparison group, and a pooled regression of REV_GROWTH_YOY on POST_GENAI, MARKET, and their interaction (firm-clustered standard errors) shows whether the post-GenAI change differs between U.S.-listed and Indian-listed firms.

Solution to RQ2

RQ2 is answered with an explanatory regression on post-GenAI firm-quarters:

$$\text{REV_GROWTH_YOY} = \beta_0 + \beta_1 \text{AI_INTENSITY} + \beta_2 \text{LOG_REVENUE} + \beta_3 \text{SGA_RATIO} + \beta_4 \text{OP_MARGIN} + \beta_5 \text{REV_GROWTH_LAG1} + \beta_6 \text{SEGMENT} + \varepsilon$$

Standard errors are clustered by firm (Petersen, 2009). H_{02} ($\beta_1 = 0$) is rejected if $p < .05$. Diagnostics include residual plots, the Breusch–Pagan test, and variance inflation factors. Fit is summarized by $R^2 = 1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$, adjusted $R^2 = 1 - (1 - R^2)(n - 1) / (n - p - 1)$, $\text{RMSE} = \sqrt{(\sum (y_i - \hat{y}_i)^2 / n)}$, and $\text{MAE} = \sum |y_i - \hat{y}_i| / n$. As a robustness check, quarter fixed effects absorb sector-wide shocks, and a random forest regressor with SHAP values (Lundberg & Lee, 2017) tests whether the AI relationship is nonlinear.

Solution to RQ3

RQ3 uses supervised machine learning to predict DECLINE_NEXT_Q from ten features: AI_INTENSITY , GENAI_MENTION , LOG_REVENUE , SGA_RATIO , OP_MARGIN , REV_GROWTH_YOY , REV_GROWTH_LAG1 , POST_GENAI , and two SEGMENT dummies. Three models of increasing flexibility are compared: logistic regression (interpretable baseline), random forest (Breiman, 2001), and gradient-boosted trees with XGBoost (Chen & Guestrin, 2016). Data are split by time: training covers 2019Q1–2023Q4, validation covers 2024, and the test set covers 2025. Hyperparameters and the decision threshold are tuned on validation only. Class imbalance is handled with class weights.

Performance on the test set is reported from the confusion matrix (true positives TP, false positives FP, true negatives TN, false negatives FN):

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad \text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad \text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

$$\text{Brier score} = (1/n) \sum (\hat{p}_i - y_i)^2$$

ROC-AUC summarizes ranking quality across all thresholds (Fawcett, 2006). H_{03} is rejected if the lower bound of the 95% DeLong confidence interval for AUC exceeds 0.50 (DeLong et al., 1988). An ablation study compares models with and without the two AI features,

using DeLong's test for correlated AUCs, to show whether AI disclosure adds predictive value beyond financial ratios. SHAP values explain which features drive predicted decline risk.

Solution to RQ4

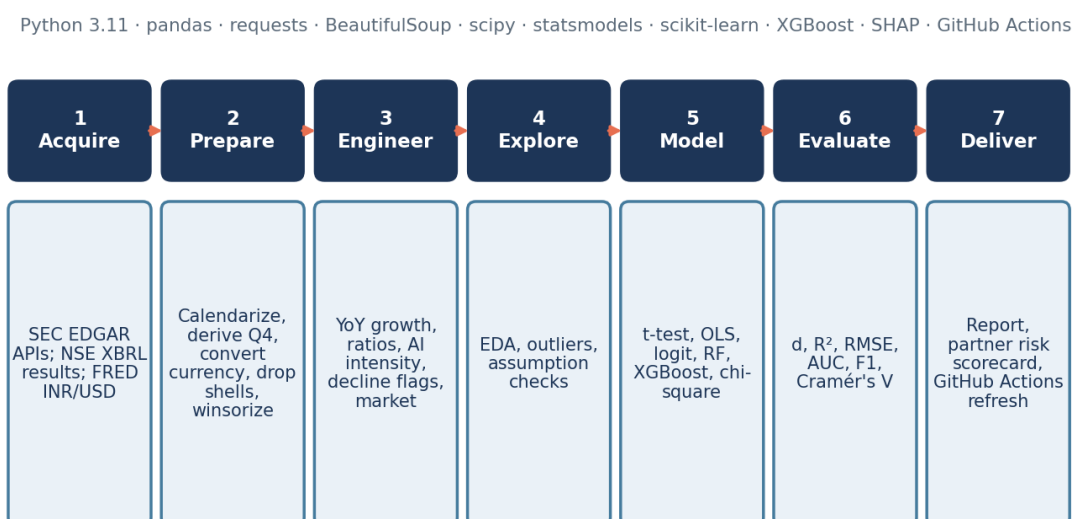
RQ4 uses a chi-square test of independence on the post-GenAI contingency table of SEGMENT (3 levels) by DECLINE_Q (2 levels):

$$\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}, \text{ where } E_{ij} = (\text{row total} \times \text{column total}) / N$$

The expected-count assumption (all $E_{ij} \geq 5$) is checked. Effect size is Cramér's $V = \sqrt{\chi^2 / (N \times (\min(r, c) - 1))}$. Standardized residuals $(O - E) / \sqrt{E}$ show which segments have more declines than expected. H_{04} is rejected if $p < .05$. A follow-up logistic regression of DECLINE_Q on SEGMENT with firm-clustered standard errors checks that the result is not driven by repeated quarters from the same firms.

Solution Pipeline and Reproducibility

Figure 4 shows the end-to-end pipeline. Data are acquired from SEC APIs, NSE XBRL filings, and FRED, prepared (calendarized, Q4 derived, Indian revenue converted to U.S. dollars, shells filtered, outliers winsorized), engineered into analysis variables, explored, modeled for RQ1–RQ4, evaluated with the metrics above, and delivered as the final report and a partner-risk scorecard. The tools are Python 3.11 with pandas, requests, BeautifulSoup, SciPy, statsmodels, scikit-learn, XGBoost, SHAP, and Matplotlib, with Git and GitHub for version control and GitHub Actions for automated runs. Reproducibility is ensured by pinned package versions, a single configuration file (*config/settings.yaml*), a fixed random seed (42), cached raw downloads, and numbered scripts that rebuild every result from source.

Figure 4*End-to-End Solution Pipeline*

Note. Each stage corresponds to a numbered script in the repository (Figure 3).

The expected outcomes are: (a) a quantified change in sector revenue growth since the GenAI inflection (RQ1); (b) an estimate of how disclosed AI emphasis relates to growth (RQ2); (c) a validated early-warning classifier with a target test AUC of at least 0.70 (RQ3); and (d) evidence of which service business models carry the highest decline risk (RQ4).

Project Plan

The work is planned in five phases over 10 weeks (Table 6). The schedule front-loads data acquisition, the main technical risk, so that any shortfall against the required sample size is found in Week 2, when the scope can still be adjusted.

Table 6*Ten-Week Project Timeline*

Week	Phase	Activities	Deliverable
1	Setup and approval	Mentor sign-off on topic; repository set-up; build firm universe; confirm available firm-quarters against N = 1,068	Approved synopsis; universe.csv
2	Data acquisition	Download XBRL company facts; extract revenue, operating income, SG&A; derive Q4; calendarize; refresh NSE comparison data	financials_quarterly.csv

Week	Phase	Activities	Deliverable
3	Data acquisition	Download 10-Q/10-K text; score AI_INTENSITY; validate dictionary on 50 sampled passages	ai_scores.csv; precision check
4	Preparation and EDA	Merge panel; clean and winsorize; finalize data dictionary; exploratory analysis and visuals	panel_firm_quarter.csv; EDA notebook
5	Statistical analysis	RQ1 t-tests and robustness; RQ4 chi-square and residuals	RQ1 and RQ4 results
6	Statistical analysis	RQ2 regression, diagnostics, fixed-effects robustness	RQ2 results
7	Machine learning	RQ3 feature pipeline; train logistic, random forest, XGBoost; tune on 2024 validation	Trained models
8	Machine learning	RQ3 test evaluation, DeLong CI, ablation, SHAP; cross-RQ robustness checks	Evaluation report
9	Synthesis	Interpret findings; build partner-risk scorecard; draft final report	Draft report; scorecard
10	Delivery	Mentor feedback; finalize report, presentation, and repository	Final submission

Recommendation and Application

The findings will be translated into recommendations for four target users.

Technology-vendor alliance and partner managers. The RQ3 model becomes a quarterly *GSI partner health scorecard* built only from public filings. Alliance teams can use it to rank partners by decline risk, focus co-selling funds and joint go-to-market investment on partners with rising AI positioning and stable growth, and hold early conversations with partners whose risk scores deteriorate. The RQ4 results show which practices (integration, application services, or managed services) to prioritize in joint offerings.

GSI leadership and strategy teams. RQ1 and RQ4 provide a benchmark of how each segment has performed since the GenAI inflection. RQ2 shows whether disclosed AI emphasis goes together with growth. If it does not, the recommendation is to shift from announcing AI to building measurable AI revenue lines. If managed services show higher decline risk, leaders should reprice outcome-based contracts before AI-driven productivity erodes effort-based revenue.

Investors and industry analysts. The panel and classifier offer a transparent, reproducible screen for identifying IT services firms whose revenue is at risk. The analysis also tests whether AI narratives in filings carry real information.

Enterprise buyers (CIOs and procurement). Segment-level evidence helps buyers anticipate which providers are under pressure and negotiate AI-driven productivity gains into renewals of labor-based contracts.

Because the pipeline runs on free public data and is automated with GitHub Actions, any of these users can refresh the results every quarter, and the Indian comparison shows whether the findings hold for India-listed GSIs as well.

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Appendix A: AI Keyword Dictionary and Named GSIs

AI_INTENSITY counts case-insensitive matches of the terms below in each 10-Q or 10-K primary document, divided by the document's word count and multiplied by 10,000. Terms marked GenAI also set GENAI_MENTION = 1. The dictionary is stored in *config/ai_keywords.txt* and will be refined after the Week 3 precision check.

Table A1

AI Keyword Dictionary

Category	Terms (regular-expression stems)
General AI	artificial intelligence; AI (as a standalone token); machine learning; deep learning; neural network; natural language processing; computer vision
Generative AI (GenAI)	generative AI; GenAI; large language model(s); LLM(s); foundation model(s); agentic; AI agent(s); copilot; ChatGPT; GPT-n

Table A2

GSIs Added by Name Outside the SIC Range

Firm	Segment assigned	Reason
Accenture plc (ACN)	Systems integration	Widely recognized GSI whose SEC SIC code lies outside 7370–7374

Note. Cognizant, DXC Technology, and Kyndryl are included automatically if their SEC SIC codes fall within 7370–7374; otherwise they are added to this list in Week 1.

Appendix B: Annotated Bibliography

The following analytical annotations cover the ten sources most central to the study's domain framing, measurement, and methods.

Babina, T., Fedyk, A., He, A., & Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics*, 151, Article 103745.

<https://doi.org/10.1016/j.jfineco.2023.103745>

The authors measure firm-level AI investment through the AI skills of employees, using resume and job-posting data for U.S. public firms, and relate it to later outcomes. Firms that invested more in AI grew faster in sales, employment, and market value, and the growth came mainly through product innovation rather than cost cutting. The study is rigorous and uses an instrumental-variable strategy, but it covers the period before generative AI and all industries rather than IT services. It supports this capstone's premise that AI positioning can relate to growth, while leaving open whether the relationship holds for labor-based services firms after 2022, which is the gap RQ2 addresses.

Brynjolfsson, E., Li, D., & Raymond, L. (2025). Generative AI at work. *The Quarterly Journal of Economics*, 140(2), 889–942. <https://doi.org/10.1093/qje/qjae044>

This field study follows more than 5,000 customer-support agents given a GenAI assistant in staggered roll-outs. Productivity, measured as issues resolved per hour, rose 15% on average, with the largest gains for less-experienced workers. The staggered design gives credible causal estimates, although the setting is a single firm and one task type. For GSIs, whose managed-services and support contracts are priced on effort, the finding implies that the same scope can be delivered with fewer hours. That is the revenue-deflation mechanism behind RQ1 and RQ4.

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702), 1306–1308.

<https://doi.org/10.1126/science.adj0998>

Using a task-level rubric applied by human raters and by GPT-4, the authors estimate the share of occupational tasks exposed to large language models. Around 80% of U.S. workers could have at least 10% of their tasks affected, with information-processing and programming occupations highly exposed. Exposure is not realized impact, and the method relies partly on model self-assessment, which the authors acknowledge. The paper motivates why IT services, built on exactly these tasks, is an important sector in which to measure GenAI effects.

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1), 35–65.

<https://doi.org/10.1111/j.1540-6261.2010.01625.x>

Analyzing a large sample of 10-K filings, the authors show that general-purpose sentiment dictionaries misclassify many words in financial text (for example, "liability" or "tax"), and they build finance-specific word lists that better predict market reactions. The paper is foundational for measuring concepts from SEC filings with dictionaries. It directly guides the construction of AI_INTENSITY in this study: a purpose-built, transparent keyword list, normalized by document length and validated on sampled passages rather than borrowed from a general lexicon.

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654), 187–192.

<https://doi.org/10.1126/science.adh2586>

In a preregistered online experiment with 453 college-educated professionals, access to ChatGPT cut the time needed for occupation-specific writing tasks by about 40% and raised output quality by 18%. The strength is clean randomization. The limitation is short, stylized tasks that may not reflect complex client projects. The study adds to the evidence that knowledge work

sold by consulting and services firms can be produced faster with GenAI, supporting the hypothesis that sector revenue dynamics shifted after 2022.

Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023). *The impact of AI on developer productivity: Evidence from GitHub Copilot* (arXiv:2302.06590). arXiv.
<https://doi.org/10.48550/arXiv.2302.06590>

In a controlled experiment, developers with access to GitHub Copilot completed an HTTP-server programming task 55.8% faster than a control group. The sample is small and the task simple, and the paper is a preprint by authors affiliated with GitHub and Microsoft, which raises a potential bias. Even so, it is among the earliest controlled measurements of AI coding assistance. Application development is a core GSI revenue line, so the paper informs this study's segment comparison (RQ4) and the expectation that application services face pricing pressure.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies*, 22(1), 435–480.
<https://doi.org/10.1093/rfs/hhn053>

Using simulations and published studies, Petersen shows that ordinary and Fama–MacBeth standard errors can be badly biased when panel data contain firm or time effects, and that standard errors clustered by firm are robust in most settings. The paper is a methodological standard in empirical finance. It justifies this study's use of firm-clustered standard errors for RQ1, RQ2, and RQ4, because repeated quarters from the same company are not independent observations.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Cohen's text defines the effect-size indices (d , f^2 , w) and conventional small, medium, and large benchmarks used in a priori power analysis, with tables and formulas for t-tests, regression, and chi-square tests. The benchmarks are generic and should ideally be replaced by domain-specific estimates, which do not yet exist for GenAI effects on IT services revenue. The

book provides the effect sizes and methods behind the sample-size calculations for RQ1, RQ2, and RQ4, and adopting "small" effects keeps the design conservative.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>

The paper introduces XGBoost, a regularized gradient-boosting algorithm with sparsity-aware split finding and efficient system design, and shows state-of-the-art results across many benchmark tasks. Its strengths are accuracy on tabular data and built-in handling of missing values, which suits financial panels where some XBRL fields are absent. The trade-off is lower interpretability, which this study offsets with SHAP explanations. XGBoost is one of the three candidate classifiers for RQ3.

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3), 837–845. <https://doi.org/10.2307/2531595>

The authors derive a nonparametric method, based on U-statistics, for estimating the variance of the area under the ROC curve and for comparing AUCs of models evaluated on the same cases. Originating in medical diagnostics, the method is now standard in machine-learning evaluation. It provides the formal hypothesis test for RQ3: a confidence interval to test AUC against 0.50, and a paired comparison to check whether adding AI-disclosure features significantly improves discrimination.